Tactile-Guided ReGrasping: Virtual Search Optimization via Extrapolated Haptic Contact

Kourosh Jolaei[1], Jean-Philippe Roberge[2], and Vincent Duchaine[2]

***Abstract*— We present a tactile-guided, local regrasp strategy that improves initially precarious grasps through proactive virtual optimization *before* physical execution. Building on shape-aware contact extrapolation, our framework extends initial tactile imprints to infer unobserved contact geometry, enabling a virtual in-plane search over $(\Delta x, \Delta y, \theta)$. Candidates are evaluated within a digital twin environment using a constrained dual-objective function that maximizes continuous haptic contact intensity while penalizing kinematic effort. We investigate a transition from structured sampling to deterministic geometric inference, introducing the Centroidal Depth Translation (CDT) paradigm. Our results, based on 602 experimental trials across canonical geometries, demonstrate that CDT identifies optimal regrasp displacements in fewer than 10 evaluations, representing a tenfold increase in search efficiency over spiral baselines. The system achieved a significant stability gain (ΔGSR $\geq 10\%$) in 90.7% of trials, consistently reaching the 85% stability plateau with a mean estimation error of only -0.7%. This approach supports proactive, low-risk haptic refinement without requiring prior object models or global vision, providing a reliable foundation for autonomous manipulation in unstructured environments.**

Keywords: Tactile Sensing; Robotic Grasping; Haptic Refinement; Virtual Search Optimization; Autonomous Manipulation; Geometric Inference

I. INTRODUCTION

Robotic grasping in unstructured environments is fundamentally constrained by incomplete or uncertain information [1]. In intelligent manufacturing and autonomous assembly, achieving high-precision manipulation remains a significant challenge, particularly when reliable visual feedback is unavailable due to clutter or occlusion [2]. In real-world scenarios, robots often encounter irregular, deformable, or previously unseen objects with uncertain properties [3]. Under such conditions, initial grasps are frequently precarious and characterized by misalignment or marginal stability. Rather than relying on a single attempt, regrasping acts as a proactive refinement step driven by sensory feedback and grasp confidence estimates [4]; many manipulation tasks inherently require more than one grasp or intermediate adjustment [5].

[1]PhD candidate, Département de génie des systèmes, École de technologie supérieure (ÉTS), Montréal, QC, Canada. mohammadtaghi.jolaeimoghaddam.1@etsmtl.ca

[2]Département de génie des systèmes, École de technologie supérieure (ÉTS), Montréal, QC, Canada. {jean-philippe.roberge, vincent.duchaine}@etsmtl.ca

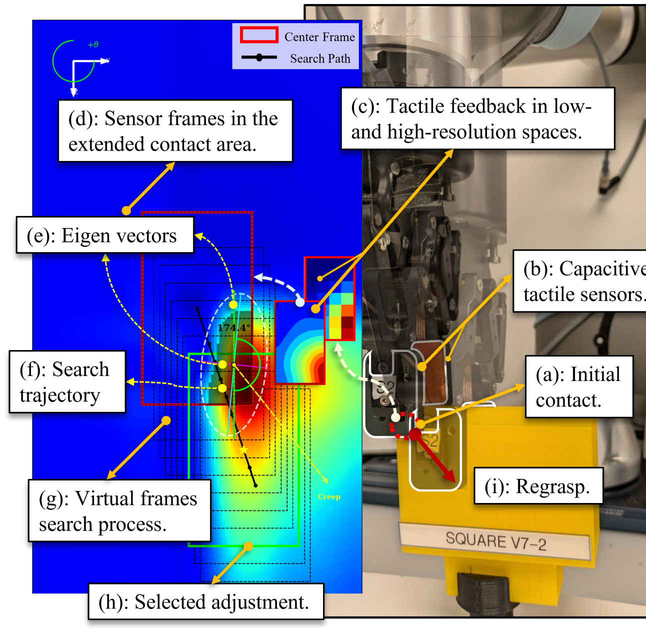

Fig. 1: Architectural overview of the tactile-guided regrasp framework. The pipeline processes initial haptic cues to extrapolate local contact geometry, synthesizing a continuous manifold for virtual optimization. This digital twin-based loop evaluates candidate adjustments to optimize grasp stability while enforcing kinematic constraints.

While vision provides essential global context, it is inherently vulnerable to lighting variations and self-occlusion during the closing phase of a grasp. Tactile sensing, by contrast, provides direct, high-fidelity contact cues that remain reliable even when the object is fully occluded by the end-effector [6]. Despite these advantages, most conventional tactile pipelines are limited to late-stage, corrective assessments, typically predicting grasp failure only moments before lift-off [7]. These systems rarely offer prescriptive guidance on how to actively navigate to a superior pose [8].

The physical implementation of regrasping, however, is fraught with risk. Large physical exploratory motions designed to remedy these poor states increase execution time, accelerate mechanical wear, and risk destabilizing the object through slippage or tipping, thereby reducing overall automation efficiency [9]. Our framework aims to mitigate these operational risks by introducing a virtual candidate-screening phase prior to physical motion, allowing the controller to predictively select configurations that optimize grasp safety with minimal displacement [10].

Unlike global, model-based grasp architectures that explore the full $SE(3)$ space using explicit object meshes, ranging from classic analytic simulators (e.g., GraspIt!) [11] to modern parallelized GPU physics environments (e.g., NVIDIA Isaac Sim) [12] or multi-view implicit neural reconstruction networks (e.g., GraspNeRF) [13], we address *tactile-guided regrasping* through a framework of localized *virtual exploration*. Upon first contact, the robot utilizes local haptic cues to reason about nearby, safer poses in a simulated environment before committing to physical movement.

The operational workflow of this proposed framework, conceptualized in Fig. 1, transitions from an initial touch to shape-aware manifold extrapolation and a subsequent virtual search to identify a stable regrasp configuration. Operationally, our architecture bridges two primary technical blocks: a latent shape-primitive classifier that extrapolates local contact manifolds, and a convolutional neural network (CNN) trained to evaluate Grasp Success Rates (GSR). By projecting the raw initial touch into this dual-component local haptic digital twin, our approach performs a highly efficient virtual search near the initial contact. We identify stability-optimized regrasp configurations through a dual-objective cost function that evaluates the trade-off between maximizing haptic contact quality and minimizing kinematic displacement, ensuring the robot commits only to motions that are both stable and physically feasible within the reachable workspace.

We evaluate two distinct search paradigms within this local framework: structured sampling and geometric inference. While structured sampling patterns, such as spirals, are robust for blind assembly tasks [14], they are inherently inefficient as they fail to utilize the available contact topology. To address this, we introduce a transition toward tactile geometric inference. By analyzing the principal eigenvectors of the initial contact spatial covariance matrix, our method deterministically calculates corrective trajectories to mitigate contact eccentricity and orientation errors. This shift from exploratory sampling to informed inference allows the system to converge rapidly on stability-enhanced configurations, significantly reducing both computational latency and the physical risk associated with large-scale search motions. The main contributions of this paper are as follows:

1) A proactive tactile-guided regrasping pipeline: We introduce a haptic digital twin framework that integrates our shape-aware contact extrapolation engine [15] with a trained Grasp Success Rate (GSR) CNN. This framework enables the robot to virtually search and evaluate various local regrasp candidates, identifying an optimized regrasp displacement.
2) Constrained dual-objective optimization for haptic refinement: We formalize a cost function optimizing the trade-off between continuous haptic contact intensity and joint-space kinematic effort, governed by multi-tier, task-space safety boundaries.
3) Extensive experimental validation and fidelity analysis: The proposed system is evaluated across 602 regrasp trials, demonstrating high correlation between virtual haptic estimates and physical grasp outcomes across diverse canonical geometries.

This paper is organized as follows: Section II contextualizes our approach within post-contact haptic decision-making. Section III details the tactile contact extrapolation engine, search paradigms, and multi-stage optimization strategy. Section IV presents a comparative analysis of search efficiency, followed by concluding remarks in Section V.

II. RELATED WORK

While contemporary vision-based grasping methodologies utilize RGB-D data, multi-view cues, or real-time 3D voxel fields to propose global $SE(3)$ grasp candidates [13], [16], [17], [18], model-based planners rely on physics simulations and analytic metrics to optimize force-closure and environmental constraints [19], [20]. Although effective in structured environments, these global strategies are constrained by a reliance on accurate object meshes or strong geometric priors, which are rarely guaranteed in unstructured scenes. Incomplete modeling frequently results in kinematic instability or unintended collisions, while the computational overhead of online replanning limits real-time adaptive capacity [21], [22]. Furthermore, visual feedback inherently degrades during the manipulation phase due to end-effector occlusion, fine-grained resolution limitations, and pose ambiguity [23], [24].

These limitations motivate local haptic strategies that operate directly at the contact interface during post-contact refinement [25]. Visuo-tactile planning frameworks have emerged to handle multi-sensory sorting and localization alignment [26], [27], [28]. While these multi-sensory architectures excel at localized state tracking, they typically treat haptic data as a secondary observation rather than utilizing a continuous, reconstructed contact manifold as the primary driver for local grasp optimization.

In contact-rich, post-contact environments where occlusion and model uncertainty prevail, regrasping is critical yet inherently high-risk; minor physical adjustments can destabilize an object, violate kinematic boundaries, or induce unintended collisions in constrained workspaces. Addressing these challenges necessitates local sensing at the contact interface to bypass unconstrained exploratory movements. Haptic feedback provides immediate, high-fidelity signals regarding pressure distribution, structural alignment, and contact evolution, enabling low-risk corrections that protect both the object and the manipulator's task space [29].

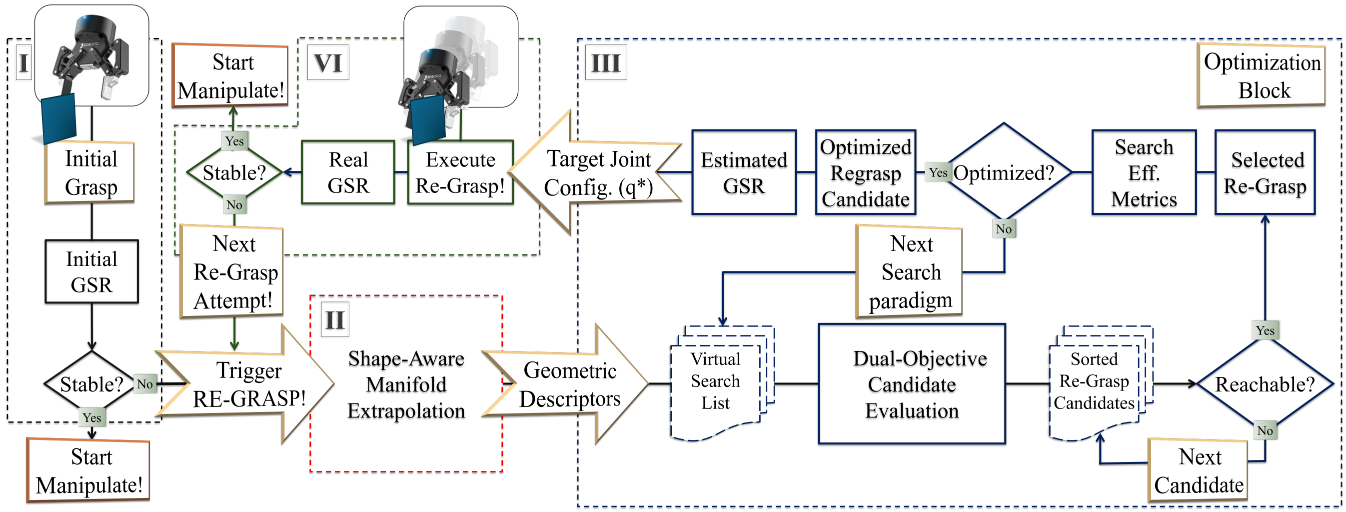

Fig. 2: Procedural pipeline of the proactive tactile-guided regrasp framework. (I) Initial touch identifies a suboptimal grasp; (II) haptic cues are utilized for shape primitive classification and manifold extrapolation; (III) a virtual in-plane search is conducted within a digital twin environment, and candidates are ranked via a dual-objective cost function to select a high-stability, low-effort displacement for physical execution.

Consequently, haptic feedback is widely utilized for post-contact grasp assessment, such as stability estimation, pre-lift success prediction, and slip monitoring [30], [31], [32], [33]. However, contemporary tactile pipelines remain largely *reactive*, acting as post-hoc diagnostic tools that classify failure modes only after a precarious state is reached. They offer minimal prescriptive guidance on the explicit corrective trajectories necessary to move to a superior configuration. This highlights the fundamental gap in haptic manipulation: the transition from passive, discrete observation to proactive trajectory guidance utilizing real-time contact geometry remains an open challenge. Unlike these discrete classification-based diagnostic approaches, our framework utilizes the continuous geometric manifold of a reconstructed tactile imprint to deterministically calculate optimal, low-risk regrasp trajectories.

To overcome these limitations, recent efforts have focused on closed-loop, proactive haptic control architectures. For instance, local grasp adjustments have been planned using sequential tactile transformations [34], iterative visuo-tactile refinement [35], and simultaneous pose estimation paired with extrinsic dexterity for in-hand manipulation [36]. Similarly, pre-execution haptic evaluation has been demonstrated in dynamic tasks [37], while model predictive control (MPC) frameworks (e.g., LeTac-MPC) and integrated perception systems (e.g., NeuralFeels) utilize real-time visuo-tactile fusion to facilitate online planning across complex contact sequences, effectively mitigating the impact of modeling errors under localized uncertainty [38], [39], [40].

While high-bandwidth reactive loops and continuous tracking architectures (e.g., TacMan-Turbo) can estimate localized geometries or object motion sequentially, they are typically limited to adjustments during the active closing phase or subject the hardware to continuous physical adjustments [41], [42]. Similarly, recent end-to-end learning architectures, such as SaTA, anchor haptic features within the kinematic frame to achieve sub-millimeter precision, but they rely heavily on mapping spatial relationships from human demonstrations [43]. In contrast to both continuous physical tracking and black-box data learning, our framework adopts an explicit optimization approach grounded in geometric inference. By utilizing the principal eigenvectors of the initial contact covariance matrix, we deterministically calculate optimal target trajectories, moving entirely beyond the uninformed search patterns common in blind manipulation pipelines.

To bridge this gap, we leverage our previously introduced shape-aware contact extrapolation framework [15] to transition from physical trial-and-error adjustments to pre-execution virtual screening. Rather than subjecting the physical hardware and object to unconstrained exploratory movements, the proposed framework evaluates local tactile manifolds within a digital twin environment, ensuring that the controller commits only to single-step physical adjustments that guarantee a more stable grasp configuration. This approach yields proactive, low-risk haptic refinement without requiring prior object profiles or CAD models, successfully complementing global vision when occluded or ambiguous [44].

III. METHODOLOGY

The primary objective of the proposed framework is to refine an initially precarious grasp by determining an optimal in-plane displacement tuple $P^* = (\Delta x, \Delta y, \theta)$ within the end-effector's task space. To achieve this, the system maps localized tactile imprints into a digital twin environment where an online optimizer searches for a target displacement that maximizes haptic contact intensity (Taxel Intensity Summation, TIS) while minimizing the joint-space kinematic effort required for the correction. This dual-objective optimization ensures that the manipulator rapidly converges onto a high-stability geometric manifold via the smallest, lowest-risk physical adjustment possible.

The complete operational workflow of this architectural pipeline is illustrated in Fig. 2, structured into three sequential operational phases:

1) Tactile Contact Extrapolation and Manifold Construction (Sec. III-A): Reconstructing discrete initial touch points into an extended, continuous 2D spatial contact map (M_{ext}) using a shape-primitive classifier (Fig. 2-II).
2) Search Paradigms: structured sampling vs. Geometric Inference (Sec. III-B): Generating potential regrasp candidates by comparing uninformed exploration trajectories against goal-oriented deterministic geometric inference paths (Fig. 2-III).
3) Constrained Dual-Objective Optimization and Execution Architecture (Sec. III-C): Resolving the active cost function trade-off ($J(P)$) under hard kinematic feasibility constraints, and executing the target trajectory on the physical robot gripper (Fig. 2-III).

A. Tactile Contact Extrapolation and Manifold Construction

Following the shape-classification methodology established in our previous work [15], the initial touch triggers a local geometric mapping sequence. The raw, discrete 7×8 tactile imprint, representing the synchronized normal pressure profile from the parallel fingertip arrays, is classified via a latent structural classifier into one of three canonical geometric shape primitives: Cuboid, Cylinder, or Sphere.

This discrete classification dictates the active, shape-specific analytical extrapolation heuristics used to reconstruct the unobserved local contact topology. The extrapolation engine expands the bounded perimeter of the initial touch, outputting a high-resolution 2D spatial contact map (M_{ext}). Rather than processing sparse, disconnected taxel points, this dense matrix models a continuous geometric manifold anchored within the finger's coordinate frame. By synthesizing the unobserved surface profile, M_{ext} establishes a smooth, continuous mathematical substrate that ensures uninterrupted gradient pathways. This allows the online optimization paradigms detailed in Sec. III-B to track spatial variations in contact intensity without encountering numerical discontinuities.

B. Search Paradigms: structured sampling vs. Geometric Inference

We evaluate two contrasting paradigms to navigate the virtual search manifold: uninformed exploration and deterministic geometric inference, with their respective trajectory profiles conceptualized in Fig. 3.

a) Structured Sampling (Blind Search): This paradigm utilizes structured, uninformed sampling patterns to identify potential regrasp candidates across the virtual manifold. Specifically, we implement two baseline trajectories: a *Square Spiral* for translational $(\Delta x, \Delta y)$ exploration, and a *Radial Spiral* that sweeps incremental in-plane rotations $(\Delta\theta)$ along the spatial sampling path. To optimize computational efficiency, these trajectories are bounded by a geometric mask derived from the shape-aware extrapolated manifold M_{ext}. By applying a binary threshold to the contact distribution, the system prunes the search space and bypasses coordinates where haptic intensity is negligible (Fig. 3(b)), avoiding the redundant execution of null "air" grasps common to unmasked baselines (Fig. 3(a)).

b) Deterministic Geometric Inference: To eliminate search latency and bypass exploratory sampling, we propose a goal-oriented *deterministic geometric inference* paradigm. Consistent with established methodologies for haptic orientation estimation via Principal Component Analysis (PCA) [45], our system extracts the principal eigenvectors of the spatial contact covariance matrix from the initial tactile footprint. By tracking these principal axes, the framework maps a direct, localized trajectory from the initial tactile centroid (TC) toward the geometric optimum (Fig. 3(c)).

Within this deterministic framework, we formalize three distinct search modes:

1) *Principal Axis Alignment (PAA):* Minimizes orientation error by centering the virtual sensor frame on the TC while simultaneously rotating the end-effector to align its longitudinal axis parallel to the major eigenvector of the contact distribution.
2) *Centroidal Depth Translation (CDT):* The virtual frame shifts deeper into the object geometry along a vector oriented at 45° relative to the principal axes while simultaneously centering the sensor frame on the TC. To preserve simulation fidelity, the maximum depth translation is bounded by a 30 mm haptic safe zone, a boundary derived from empirical limits where extrapolated contact profiles correlate tightly with physical reality [15].
3) *Unified Geometric Refinement (UGR):* Combines these paradigms into a single pass, executing a hybrid adjustment that simultaneously optimizes for translational centering, orientational alignment, and the 45° depth pursuit within the 30 mm safe boundary.

Rather than executing a single unverified leap upon calculating these trajectories, the target path is discretized into a finite sequence of intermediate steps. This discretization allows the system to implement a high-fidelity *reverse-stepping evaluation strategy*: the optimizer first validates the terminal goal displacement; if it violates the kinematic or workspace boundaries

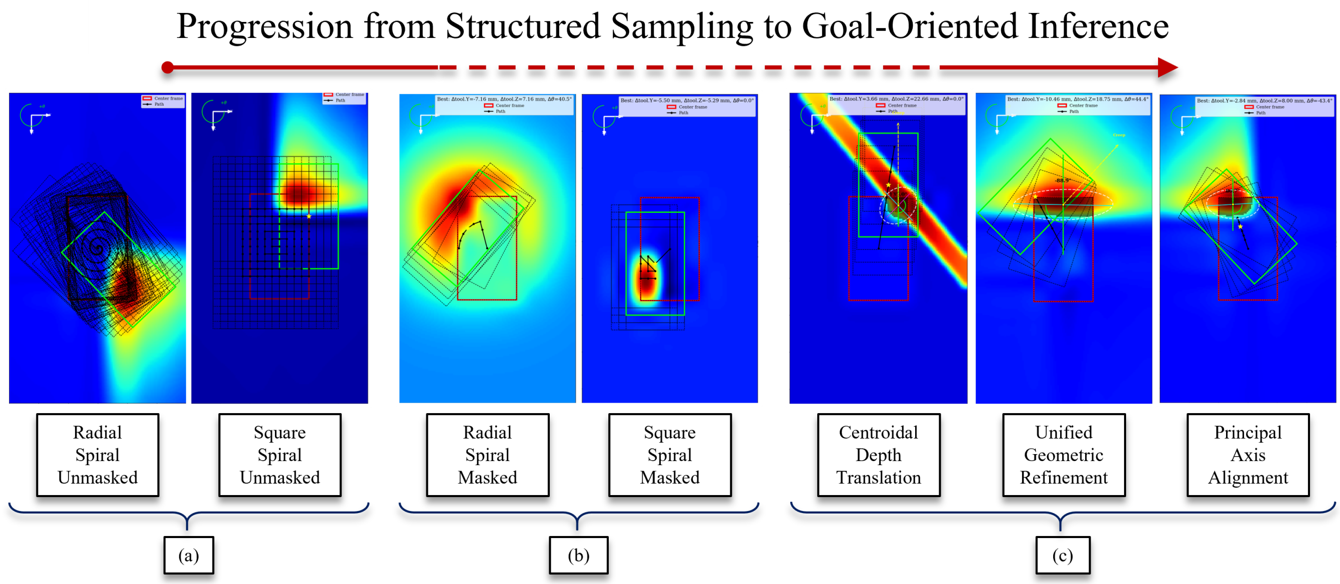

Fig. 3: Comparative visualization of virtual search strategies over an extrapolated contact manifold. (a) structured sampling (Unmasked): Exhaustive spiral sampling regardless of haptic probability. (b) structured sampling (Masked): Manifold-aware pruning bypasses regions of zero haptic intensity to improve efficiency. (c) Deterministic Geometric Inference: Direct trajectory calculation via principal eigenvectors, targeting the stability optimum to minimize search latency. For all strategy heatmap overlays, the red frame denotes the original sensor frame at the initial touch state, the dotted black frames track the sequential candidate search steps, the central solid black line shows the progressive search trajectory path, and the green frame isolates the optimal candidate adjustment selected for physical realization.

defined in Sec. III-C, the framework iteratively steps backward to evaluate the preceding intermediate candidate. By integrating this strategy, these modes ensure that the manipulator rapidly converges on the highest-fidelity target displacement that remains physically feasible, maximizing proximity to the geometric optimum while fundamentally reducing the physical risk of object destabilization.

C. Constrained Dual-Objective Optimization and Execution Architecture

A candidate displacement tuple $P = (\Delta x, \Delta y, \theta)$, representing the relative adjustment from the initial contact state, is selected for physical execution only if it satisfies a sequential cascade of hard kinematic safety gates and minimizes a constrained dual-objective cost function $J(P)$. Rather than allowing competing geometric goals to conflict within the active search loop, the framework treats task-plane alignment, joint limits, and singular configurations as non-negotiable pruning boundaries. In particular, out-of-plane velocities are explicitly suppressed via rigid task-plane boundaries. This structural separation ensures that the continuous optimization loop is restricted entirely to evaluating the direct trade-off between maximizing continuous haptic contact intensity and minimizing joint-space configuration changes.

a) Dual-Objective Cost Function: The cost function is parameterized by a set of weighting coefficients and saturation caps, allowing the optimization behavior to be tuned according to the specific requirements of the manipulation task. $J(P)$ is defined as a pure dual-objective formulation:

$$J(P) = \lambda_q \cdot C_{\text{quality}} + \lambda_e \cdot C_{\text{effort}} \tag{1}$$

where λ_q and λ_e are weighting coefficients that define the relative importance of haptic stability versus motion efficiency, allowing the optimizer to be tuned to specific task requirements.

Contact Quality (C_{quality}) rewards configurations that maximize haptic engagement, defined analytically as:

$$C_{\text{quality}} = (1 - \text{TIS}(P))^2 \tag{2}$$

here, $\text{TIS}(P)$ represents the normalized Taxel Intensity Summation, which aggregates the predicted normal pressure across the extrapolated manifold M_{ext}. This value is normalized relative to the maximum theoretical intensity achievable within the shape-specific safe zone, ensuring the cost remains bounded within $[0, 1]$.

We deliberately utilize TIS as the primary optimization metric rather than a Grasp Success Rate (GSR) network based on a strategic separation between active trajectory guidance and final state assessment. As demonstrated in contemporary tactile-based grasp stability research [46], localized intensity metrics, specifically tactile pressure footprints, exhibit a strong, direct correlation with physical grasp safety limits, providing a more robust substrate for continuous optimization than saturated binary classification metrics. Crucially, while a binary or sigmoid-saturated GSR classifier generates flat plateaus and steep non-differentiable boundaries that choke mathematical search frameworks, the raw surface pressure aggregation of TIS provides a continuous, highly linear gradient pathway across the contact manifold. This smooth substrate enables the online optimizer to track fine-grained candidate variations effectively without getting trapped in local minima or experiencing gradient vanishing.

Consequently, TIS is deployed exclusively to drive the pre-execution virtual screening loop, leaving the non-linear GSR network to function as a decoupled, high-level evaluative proxy. This separation underpins the multi-stage performance metrics detailed in Sec. IV, including absolute GSR gain, convergence thresholds, and the tactile return-on-investment profile (A_{tis}), ensuring our control law optimizes via continuous geometric primitives while strictly validating outcomes through absolute success metrics.

Kinematic Effort (C_{effort}) promotes motion efficiency and minimizes mechanical wear. The kinematic effort is defined by the squared Euclidean (L_2) norm of the projected joint-space displacement vector $\Delta\mathbf{q}_{\text{opt}}$:

$$C_{\text{effort}} = \|\Delta\mathbf{q}_{\text{opt}}\|_2 \tag{3}$$

This quadratic formulation provides a continuous, strictly convex penalty on joint-space displacement, ensuring that the optimizer naturally prioritizes motion efficiency while maintaining a stable gradient throughout the entire search space. This design choice ensures that candidates are evaluated based on the direct trade-off between haptic engagement and motion expenditure, with final candidate viability subsequently guaranteed by the reachability filters defined in the following paragraphs.

b) Sequential Reachability Filtering: To ensure operational safety and hardware integrity, each candidate displacement P must satisfy a three-tier feasibility hierarchy. Adjustments that fail any tier are discarded immediately, preventing unfeasible adjustments from corrupting the optimization loop:

- **Workspace and Rigid Task-Plane Constraints:** Candidates are strictly confined to the Oriented Bounding Box (OBB) of the extrapolated contact manifold M_{ext}. To suppress out-of-plane tracking errors and drift, candidates are explicitly projected onto the contact plane, restricting the search space to a rigid three-DOF task $(\Delta x, \Delta y, \theta)$. Furthermore, the search envelope is bounded by operational thresholds for maximum translation ($R_{\max}$) and rotation ($\Theta_{\max}$) to preserve the locality of the refinement.
- **Singularity Avoidance:** To ensure controllable end-effector velocities and avoid kinematic singularities, the system evaluates the condition number $\kappa(\mathbf{J})$ of the manipulator Jacobian. Adjustments are rejected if $\kappa(\mathbf{J}) > \kappa_{\max}$, ensuring that the selected regrasp remains within a well-conditioned region of the robot's workspace where inverse kinematics (IK) solutions are robust.
- **Configuration Space ($\mathcal{C}$-space) Limits:** The final tier enforces the physical boundaries of the UR5e manipulator. A candidate is discarded if it requires a joint configuration $\mathbf{q}$ that exceeds hardware position limits or results in a step-wise displacement exceeding a safety-critical threshold $\Delta\mathbf{q}_{\max}$.

c) Kinematic Realization and Constrained Execution: While MoveIt2 provides the high-level infrastructure for global inverse kinematics and collision-free path planning, the precise local refinement of the regrasp requires a more granular, manifold-tangent control law. Therefore, we supplement the standard software stack with a custom resolved-rate execution module.

To translate the optimal virtual candidate $P^* = (\Delta x^*, \Delta y^*, \theta^*)$ into physical motion, the system maps the in-plane displacement to a spatial twist $\mathbf{v} \in \mathbb{R}^6$ at the end-effector. In our coordinate convention, Δx and Δy are aligned with the gripper's `tool0.Y` and `tool0.Z` axes, while the rotation θ is performed about the tactile sensor's geometric center via a pivot-aware transformation. This mapping accounts for the physical lever arm (L) and width offset (W) of the gripper, ensuring that orientational adjustments occur around the contact manifold rather than the robot's wrist center. We compute the realized joint-space increments, $\Delta\mathbf{q} \in \mathbb{R}^6$, using a Damped Weighted Least-Squares (DWLS) approach:

$$\Delta\mathbf{q} = (\mathbf{J}^\top\mathbf{W}\mathbf{J} + \lambda^2\mathbf{I})^{-1}\mathbf{J}^\top\mathbf{W}\mathbf{v} \tag{4}$$

where $\mathbf{J}$ is the manipulator Jacobian, $\mathbf{W}$ is a diagonal weighting matrix heavily biased to mathematically suppress out-of-plane velocities and prioritize task-plane tracking accuracy, and λ is a damping factor to ensure numerical stability near singularities. To maintain strict adherence to the local finger-plane and prevent trajectory drift, the Jacobian $\mathbf{J}$ is re-evaluated and updated iteratively at every control segment ($n = 1$). This continuous linearization allows the system to suppress non-linear integration drift throughout the execution phase.

To minimize Cartesian tracking error, the control law incorporates a Proportional leash controller ($K_p = 0.7$). This term computes corrective velocities based on the error between the physical tool pose and the ideal interpolated path, which are integrated into the spatial twist $\mathbf{v}$ before the DWLS inversion.

Prior to motion, the gripper is opened to break static friction (stiction) and decouple the sensor surface from the object, thereby preventing lateral shear damage and unintended object tipping. The trajectory is dynamically discretized into N segments, defined by a 0.4 mm linear and 0.2° angular resolution, maintaining the system in a quasi-static regime to minimize dynamic disturbances. Each step is verified against a maximum displacement threshold $\Delta\mathbf{q}_{\max}$ and the Jacobian condition limit $\kappa_{\max}$. Upon arrival at the target displacement, the gripper re-engages the object, initiating a secondary haptic acquisition phase to verify the final improvement in grasp stability.

IV. RESULTS AND DISCUSSION

To validate the efficacy of the proposed proactive haptic regrasp framework, we conducted a comprehensive empirical evaluation across a dataset of 15 diverse physical objects. These test cases were selected to populate the three canonical geometric shape primitives established in our perception pipeline. The system was intentionally initialized with 86 distinct precarious grasping configurations characterized by marginal initial stability or severe spatial misalignment. Across these configurations, we executed a total of 602 physical regrasp trials, providing a statistically robust foundation to compare uninformed sampling patterns against deterministic geometric inference modes.

A. Experimental Setup

The experimental setup consists of a UR5e robotic arm equipped with a Robotiq 2F-85 parallel-jaw gripper. Each finger is instrumented with a multimodal capacitive tactile sensor developed by the CoRo Lab [47]. Each sensor features a 7×4 array (28 taxels) for high-resolution normal pressure sensing at 60 Hz and integrates an IMU (gyroscope, accelerometer, magnetometer) to estimate finger orientation relative to the palm. When the parallel fingertips engage an object, their dual arrays provide the synchronized base contact footprint mapped by the extrapolation layers. Static pressure profiles are sampled via a transimpedance amplifier at 1 kHz, establishing the baseline normal force distribution used directly for spatial manifold reconstruction and online optimization.

The control architecture is hosted on an Ubuntu 22.04 LTS workstation, interfaced with the UR5e controller via a low-latency Ethernet connection. The software stack leverages the Robot Operating System 2 (ROS2) Humble ecosystem for synchronized inter-process communication, with MoveIt2 serving as the primary infrastructure for inverse kinematics (IK) mapping and collision-free global path planning. All high-level algorithmic layers, including real-time tactile stream acquisition, shape-aware manifold extrapolation, and the virtual search optimizer are implemented as modular Python 3.10 nodes within this framework.

B. Haptic Quality Enhancement and Search Fidelity

The primary objective of the local virtual search layer is to proactively discover target geometries that maximize post-contact grasp stability before physical movement occurs. Fig. 4 and Fig. 5 summarize the empirical performance and tracking accuracy of the proposed paradigms across all experimental trials.

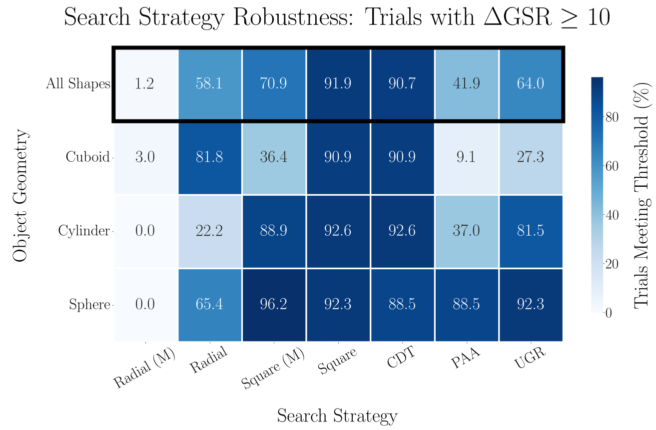

Fig. 4: Robustness matrix illustrating the percentage of trials achieving a significant grasp success improvement (ΔGSR $\geq$ 10%) categorized by object geometry and active search method.

To evaluate the operational robustness of each search strategy, we first analyze the frequency with which significant stability gains are achieved, defined as the percentage of individual trials yielding a net improvement of ΔGSR $\geq 10\%$. As illustrated in the robustness matrix (Fig. 4), the proposed deterministic Centroidal Depth Translation (CDT) demonstrates exceptional geometric reliability, with 90.7% of all trials across all combined shape classes successfully exceeding this performance threshold. Among the baseline sampling methods, the unmasked Square Spiral strategy achieved a comparable frequency of 91.9%, while its haptic-masked counterpart, Square (M), achieved 70.9%. Conversely, purely orientation-driven adjustments like Principal Axis Alignment (PAA) fell to 41.9%, demonstrating that translational adjustments to clear contact eccentricity are primary drivers for stabilizing precarious initial touches.

A core contribution of our framework is the predictive fidelity of the shape-aware contact extrapolation engine. Fig. 5 (Top) tracks the evolution of the "GSR Shift" across three sequential operational stages: the Initial touch state ($\mathcal{A}$), the Virtual Search candidate prediction ($\mathcal{V}$), and the final physical verification outcome ($\mathcal{P}$). The absolute tracking error between virtual optimization and physical reality is explicitly quantified by the red-boxed metrics at the top of the plot. For our high-reliability methods, this absolute discrepancy remains remarkably low: -0.3% for Square (M), $+0.2\%$ for unmasked Square, and $+2.1\%$ for CDT. Across all evaluated paradigms, the framework exhibits a minimal global mean estimation error of only -0.7%, confirming that the digital twin environment accurately models real-world physical boundaries.

As shown in the net gain distributions (Fig. 5, Bottom), the unmasked Square Spiral and the deterministic CDT modes achieved the highest absolute performance improvements, yielding individual mean gains of $+35.4\%$

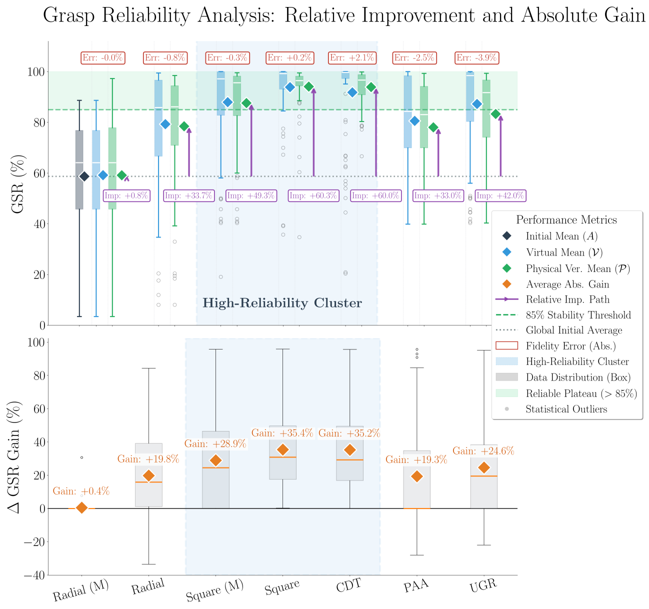

Fig. 5: Multi-stage grasp reliability and system fidelity analysis. (Top) Empirical distribution of GSR values across the framework pipeline: Initial precarious state ($\mathcal{A}$), Virtual Search candidate prediction ($\mathcal{V}$), and physical Verification Mean ($\mathcal{P}$). Red boxed labels quantify the absolute systemic Fidelity Error ($|\mathcal{P}-\mathcal{V}|$). Purple arrows indicate relative tracking improvement paths, while the green dashed line marks the 85% stability threshold. (Bottom) Absolute net GSR gain (ΔGSR) per strategy, revealing a high-reliability cluster composed of CDT, Square, and Square (M) variants.

and +35.2%, respectively.

Crucially, the deterministic CDT and Square variants formed a highly consistent *high-reliability cluster* (shaded region, Fig. 5), the only strategies capable of lifting the manipulator out of its precarious initial baseline (58.68% GSR) and past the critical 85% stability threshold. This 85% target boundary is grounded in the foundational findings of Kwiatkowski *et al.* [48], which identified a strict performance *plateau* near this level due to the inherent limits of data-driven tactile classifiers. Rather than attempting to break through this statistical ceiling via exhaustive network training, our framework leverages shape-aware contact extrapolation to proactively guide the end-effector directly into this optimal zone, effectively translating a precarious touch into a configuration where grasp stability is functionally guaranteed.

C. Search Efficiency and Computational Performance

While multiple paradigms can successfully identify high-quality grasp configurations, their operational utility is strictly defined by search efficiency, computational overhead, and outcome reliability. We evaluate these core characteristics through a comprehensive multi-metric comparative analysis, contrasting uninformed sampling against our deterministic geometric inference modes: Principal Axis Alignment (PAA), Centroidal Depth Translation (CDT), and Unified Geometric Refinement (UGR).

To measure the absolute magnitude of the haptic stabilization, we define the **Total GSR Gain** (ΔGSR):

$$\Delta\text{GSR} = G_{\text{final}} - G_{\text{initial}} \tag{5}$$

where G_{initial} and G_{final} represent the physical Grasp Success Rate (GSR) measured before and after the regrasp execution, respectively. This metric serves as the primary indicator of search effectiveness, providing the baseline improvement upon which the efficiency and reliability metrics are calculated.

A critical differentiator between the paradigms is the GSR Convergence Rate (η_{gsr}), which quantifies the stability gain achieved per unit of computational evaluation effort:

$$\eta_{\text{gsr}} = \frac{\Delta\text{GSR}}{N_{\text{eval}}} \tag{6}$$

where N_{eval} represents the exact number of virtual forward passes through the scoring network required to identify the stability optimum.

To assess the consistency of the controller across diverse object geometries, we evaluate the Outcome Variance (σ_{final}), representing the standard deviation of the final physical GSR across all experimental runs:

$$\sigma_{\text{final}} = \sqrt{\frac{1}{K}\sum_{k=1}^{K}(G_{\text{final},k} - \bar{G}_{\text{final}})^2} \tag{7}$$

where K is the total number of unique object-pose trials, $G_{\text{final},k}$ is the physical outcome of trial k, and $\bar{G}_{\text{final}}$ is the mean success rate for that specific strategy. To facilitate a consistent multi-dimensional comparison where higher values uniformly represent superior performance, we report the *Outcome Stability* in our evaluations, defined as the complement of the normalized standard deviation $(1 - \hat{\sigma}_{\text{final}})$.

The mechanical cost of a physical regrasp is quantified by the Kinematic Efficiency (E_{kin}), representing the stability gain normalized by the total joint-space displacement:

$$\text{E}_{\text{kin}} = \frac{\Delta\text{GSR}}{\|\Delta\mathbf{q}\|_2} \tag{8}$$

where $\|\Delta\mathbf{q}\|_2$ is the L_2 norm of the joint-space displacement vector. High E_{kin} values prioritize economical, low-risk adjustments that reach the target stability plateau with minimal physical reconfiguration of the arm.

Finally, we evaluate the geometric efficiency of the haptic adjustment through the TIS Alignment metric (A_{tis}). Rather than measuring the absolute magnitude of success, this metric functions as an economic return-on-investment profile, quantifying the ratio of physical

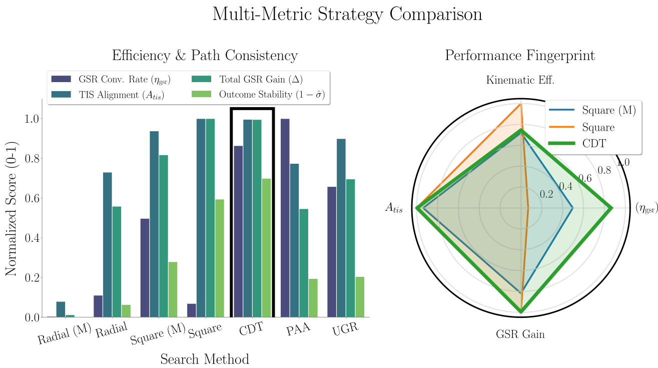

Fig. 6: Multi-metric performance evaluation. (Left) Efficiency and path consistency breakdown. Bars represent normalized scores scaled between $[0, 1]$ across four indicators: GSR convergence rate (η_{gsr}), TIS alignment (A_{tis}), total GSR gain (Δ), and outcome stability ($1-\hat{\sigma}$). The black frame isolates the Centroidal Depth Translation (CDT) to highlight its superior balance across the multi-metric performance envelope. (Right) Strategy performance fingerprint of the high-reliability cluster via a polar radar chart, illustrating the physical trade-off between haptic alignment, kinematic efficiency, and absolute tracking stability.

stability improvement relative to the scaled change in contact engagement:

$$A_{\text{tis}} = \frac{\Delta \text{GSR}}{\alpha \cdot (\text{TIS}_{\text{final}} - \text{TIS}_{\text{initial}})} \tag{9}$$

where $\text{TIS}_{\text{initial}}$ and $\text{TIS}_{\text{final}}$ represent the physical Taxel Intensity Summation values recorded before and after execution, respectively, and $\alpha = 0.01$ is a scaling constant utilized to normalize haptic intensity relative to the stability gain.

A high A_{tis} score indicates a highly economical strategy that successfully maximizes grasp stability by tracking the continuous geometric features of the extrapolated manifold, rather than relying on an unconstrained increase in normal force.

While the raw TIS value serves directly as the continuous cost substrate inside the virtual objective function $J(P)$, the A_{tis} metric evaluates the physical realization of that selection, verifying that real-world stability gains are driven by superior geometric alignment.

D. Multi-Metric Strategy Comparison

A critical differentiator between the virtual search paradigms is the resulting GSR Convergence Rate (η_{gsr}), as visualized in the multi-metric efficiency breakdown (Fig. 6, Left). While the spiral baselines exhibit heavily degraded convergence scores due to the unguided, exhaustive nature of their spatial sampling loops, the deterministic geometric inference modes (PAA, CDT, and UGR) demonstrate an inherent goal-oriented efficiency. By utilizing the principal axes of the contact covariance matrix to compute direct trajectories, the deterministic modes bypass exploratory latency entirely. As shown by the normalized performance indicators, the high-reliability cluster, composed of CDT, Square, and Square (M) variants, maintains near-peak capabilities in total GSR Gain (Δ), verifying that these adjustments successfully push the manipulator past the target stability threshold.

However, raw gain alone does not define operational excellence. The TIS Alignment (A_{tis}) scores in Fig. 6 reveal that the CDT paradigm achieves its stability through superior geometric alignment. A high A_{tis} score indicates that the strategy precisely utilizes the extrapolated manifold to find an optimal contact. This suggests that the shape-aware engine provides the robot with a sufficient topological information to facilitate proactive, highly economical regrasp trajectories without requiring prior object models or risking structural damage from over-squeezing.

The systemic robustness of these strategies across diverse geometric primitives is further validated by the inverted Outcome Stability ($1 - \hat{\sigma}$). As illustrated in the comparative bar chart, the CDT, Square, and Square (M) methods yield the highest scores for final outcome consistency, indicating tightly bounded post-contact success distributions across all canonical shapes. This multi-dimensional balance is elegantly captured by the Strategy Performance Fingerprint (Fig. 6, Right). While the Square Spiral (Orange) prioritizes raw Kinematic Efficiency (E_{kin}) at the expense of absolute GSR Gain magnitude, and the haptic-masked Square (M) variant (Blue) focuses on maximizing absolute GSR Gain, the deterministic CDT mode (Green) occupies the most balanced envelope of the Pareto-optimal space. By tracking the principal contact eigenvectors, CDT simultaneously maximizes real-world stability gains, ensures optimal haptic alignment economy, and preserves computational speed.

E. Outcome Reliability and the Impact of Rotation

Beyond raw performance gains, the statistical consistency of the final grasp configuration serves as a direct proxy for controller reliability. As demonstrated across all 602 experimental trials, strategies that incorporate rotational degrees of freedom ($\Delta\theta$), namely Radial, PAA, and hybrid UGR, suffer from a severe dual-stage performance penalty. This degradation originates directly within the virtual optimization layer; because in-plane rotations swing portions of the parallel fingertip arrays outside shape-specific haptic safe zones, the continuous cost function $J(P)$ actively penalizes these candidates due to localized drops in predicted contact intensity.

Consequently, the optimized Virtual Means ($\mathcal{V}$) for PAA (80.56%) and UGR (87.25%) are inherently bounded below the elite baselines established by translation-only paradigms like CDT (91.83%), completely independent of physical execution dynamics.

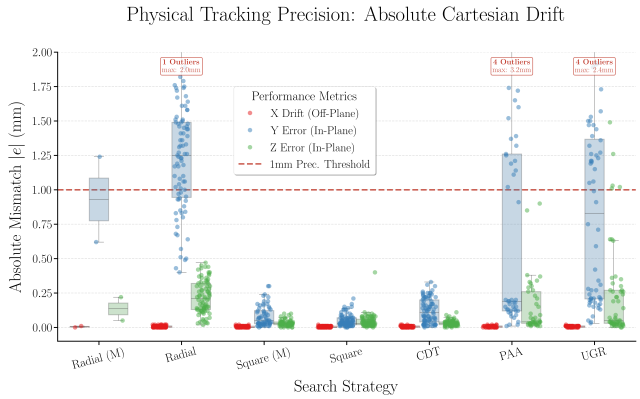

Fig. 7: Physical tracking precision and Cartesian mismatch profiles across the seven search strategies. The absolute tracking error ($|e|$) is explicitly decomposed into off-plane drift (X) and task-space in-plane slide (Y, Z). Across 602 total physical regrasp trials involving corrective displacements up to 32.13 mm ($\mu = 9.27$ mm, $\sigma = 8.47$ mm), the DWLS inverter successfully suppresses off-plane drift to sub-millimeter levels. However, the introduction of rotational terms in the Radial, PAA, and UGR strategies results in higher in-plane tracking error variance.

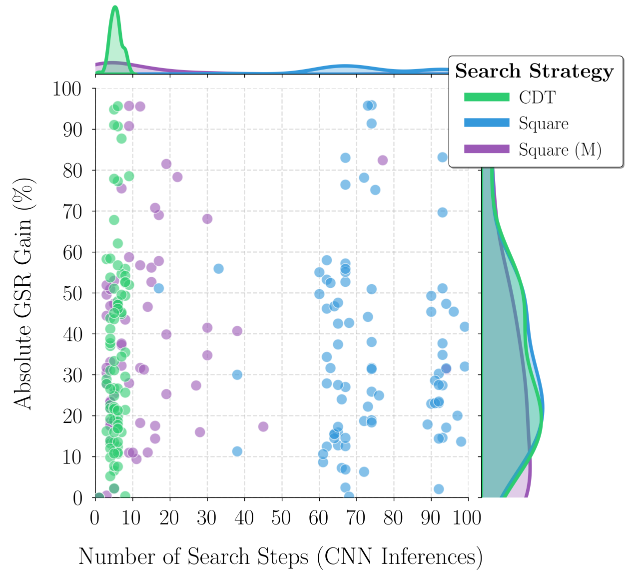

Fig. 8: Computational Efficiency vs. Stability Gain. Distribution of search trials on the Pareto Frontier. structured sampling methods (spirals) exhibit high inference counts, while deterministic geometric inference (CDT) achieves significantly higher ΔGSR gains with minimal computational overhead.

This simulation-level constraint is subsequently compounded by real-world physical tracking anomalies during execution. As illustrated in the Cartesian error profiles (Fig. 7), methods involving orientation adjustments exhibit in-plane errors (e_y, e_z) nearly an order of magnitude higher than translational workflows. When the physical manipulator executes coupled multi-DOF configurations, orientational updates induce severe unmodeled tactile shear forces and localized surface slipping. This mechanical uncertainty accounts for the notable vertical outcome dispersion and degraded physical verification means ($\mathcal{P}$) seen in Fig. 5 (78.038% for PAA, 83.3% for UGR).

While the DWLS inverter successfully suppressed off-plane drift (X) to strict sub-millimeter levels across all trials, the combination of initial optimization limits and high physical tracking variance confirms the physical risk of rotational maneuvers. These findings reinforce the conclusion that prioritizing translation-only alignment via deterministic geometric inference provides a safer, more predictable, and structurally superior pathway for autonomous post-contact grasp refinement.

F. *Identification of the Multi-Metric Winner*

To identify the most robust operational strategy, we analyze the performance characteristics across the defined metric suite (Fig. 6). While the UGR and PAA methods provide high convergence rates (η_{gsr}), the Centroidal Depth Translation (CDT) emerges as the optimal paradigm when balancing haptic alignment against outcome stability. As visualized in the Strategy Performance Fingerprint (Fig. 6-Right), CDT maximizes the integrated search utility by maintaining a superior balance across all dimensions. Specifically, it achieves a high raw stability gain (ΔGSR) while simultaneously maximizing kinematic efficiency (E_{kin}). This high outcome consistency, reflected in the condensed interquartile ranges in Fig. 5 (Top), confirms that CDT provides a more repeatable path to the 85% stability plateau than structured sampling or rotation-heavy alternatives.

The relationship between computational effort and haptic improvement is captured in the Search Pareto Frontier (Fig. 8). Spiral patterns exhibit high search complexity, often requiring 60 to 100 CNN inferences to identify a local optimum. In contrast, the deterministic inference modes cluster in the Pareto-optimal *high-gain, low-latency* quadrant, typically reaching the optimal candidate in fewer than 10 evaluations. This represents a tenfold increase in search efficiency, effectively minimizing the decision-making bottleneck in the regrasping pipeline.

Consequently, we introduce CDT as the winning paradigm for shape-aware proactive grasping; it provides the *"best of both worlds"*: the decision-making speed of a deterministic geometric inference and the execution reliability of a translation-only movement.

V. Conclusion

In this work, we presented a proactive, tactile-guided regrasp framework that utilizes shape-aware contact extrapolation to optimize grasp stability prior to physical execution. By shifting the search process into a virtual digital twin environment, our approach significantly reduces the mechanical risks and computational bottlenecks associated with traditional exploratory motions.

Our empirical evaluation of 602 trials demonstrates that deterministic geometric inference, specifically the Centroidal Depth Translation (CDT) paradigm, provides a superior operational balance. CDT achieved a significant GSR gain (ΔGSR $\geq$ 10%) in 90.7% of trials while consistently reaching the 85% stability plateau. Crucially, by utilizing the principal eigenvectors of the tactile distribution, CDT provided a tenfold reduction in computational latency compared to spiral alternatives, identifying optimal candidates in fewer than 10 CNN inferences. These results suggest that local contact cues, when properly extrapolated, are sufficient to drive high-fidelity proactive planning without requiring prior object models or global vision.

In the current work, the focus was on robotic regrasping for rigid objects, where the contact manifolds were modeled based on static tactile impressions and virtual contact projections. Future research will explore the incorporation of deformable objects into the regrasping pipeline. This will necessitate the development of dynamic, shape-aware contact models capable of adapting in real-time as the object geometry evolves under load. Introducing compliance adds a significant layer of complexity to the stability manifold, requiring more sophisticated predictive models to manage the coupling between deformation and grasp success. This extension will enhance the robustness of the regrasping system in more varied and realistic environments, where object shapes and contact points change during manipulation.

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